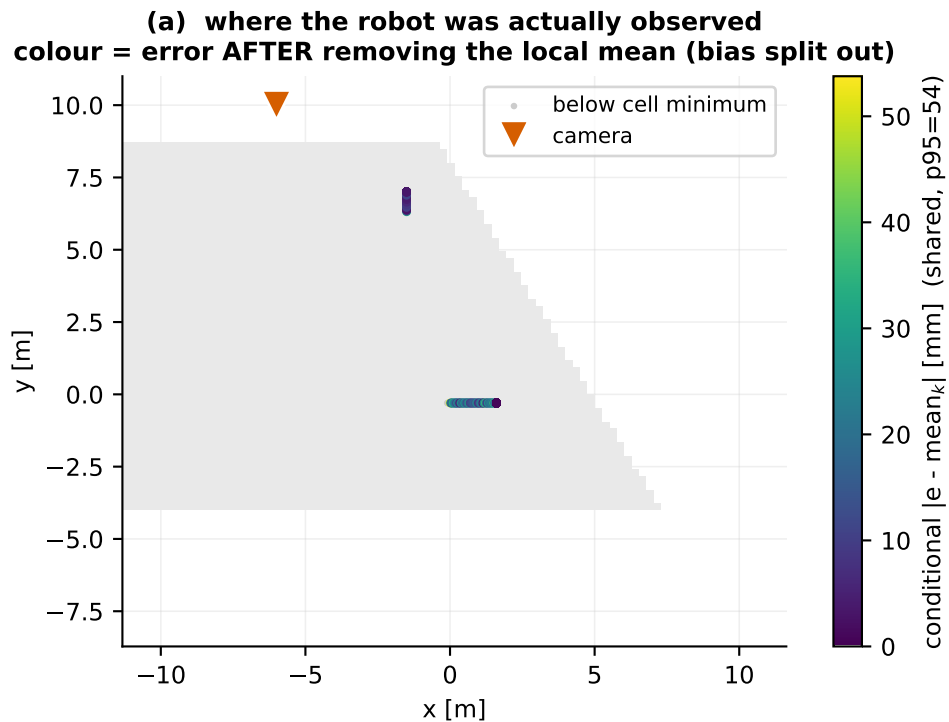
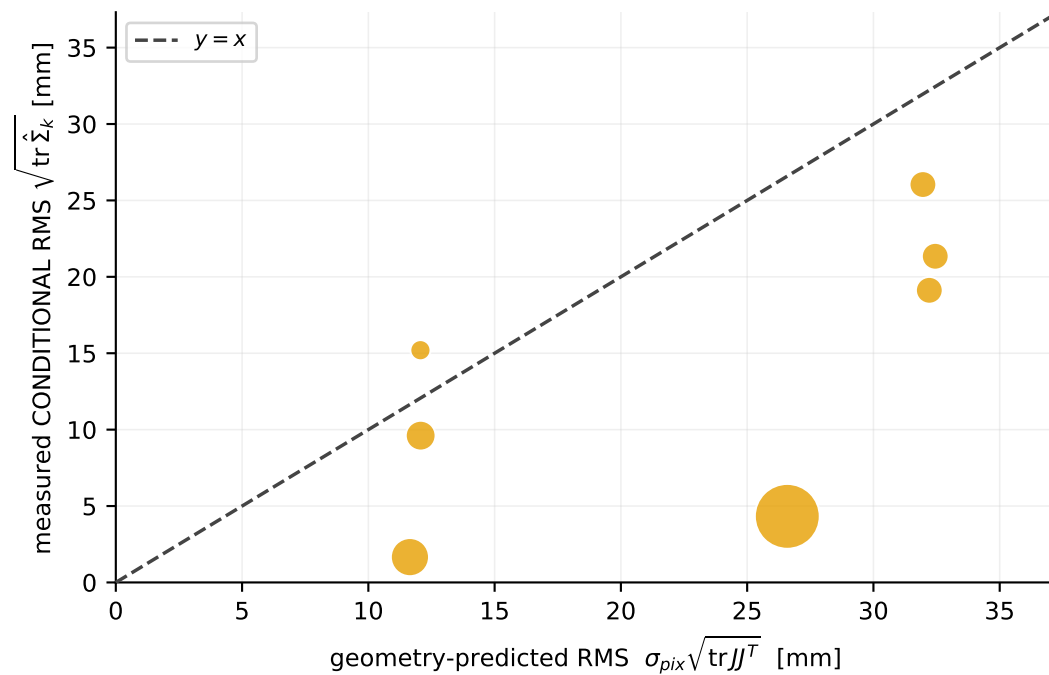


Camera B — deployed-corrected residual



(b) does geometry predict the local SPREAD?
N = 295 detections · M = 7 cells (0.5 m, ≥ 5 each)
conditional: R^2 vs identity = -0.91 · cell Spearman = +0.75 · median obs/pred = 0.66
median per-cell BIAS removed first = 20 mm
ONLY 7 CELLS — too few to establish a trend; read as a scale check, not a fit



Point area in (b) scales with detections in the cell. σ_{pix} is fitted on leave-region-out folds, so each cell's prediction is out-of-region. Ground truth positions the dots in (a) and measures the residual; it never enters a projection, Jacobian or covariance. **READ THE LIMIT BEFORE THE RESULT:** only 3.1 x 7.3 m is sampled and image column tracks range at $|\rho| = 0.73$, so range, image row and projection conditioning are collinear here. A weak or negative R^2 does NOT show that geometry fails -- it shows this dataset cannot test it.